

Machine Learning A (2026)

Home Assignment 2

Your name and student ID

Contents

1 Preprocessing (35 points)

1.1 Importance of Preprocessing (8 points)

1.2 Input Centering (9 points)

1.2.1 (a)

1.2.2 (b)

1.3 Input Whitening (18 points)

1.3.1 (a)

1.3.2 (b)

1.3.3 (c)

1.3.4 (d)

2 Hoeffding's Bound (15 points)

3 Illustration of Markov's, Chebyshev's, and Hoeffding's Inequalities (25 points)

3.a

1.

2.

3.

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Figure 1: Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris.

3.b [optional, 0 points]

```
# Example code
# Creating an example array
data = np.array([5, 2, 8, 1, 6])

# Calculating cumulative sum using cumsum
cumulative_sum = np.cumsum(data)
```